AGENTIC AI FOUNDATION · AAIF

# Update on the AAIF
# *Observability & Traceability Working Group*

What the AAIF's newest working group is doing, and where it may need IETF contributions

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# A new working group in a new foundation

**The Agentic AI Foundation (AAIF)** is the Linux Foundation home for open agentic-AI infrastructure, including the Model Context Protocol (MCP) and the Goose framework. The Observability & Traceability WG (OT-WG) is one of its first chartered working groups.

**Chartered spring 2026**

Ratified 15 April 2026 · approved 13 May 2026.

**Chairs**

Pavan Sudheendra (Cisco, Chair) & Matthew Lee (Datadog, Co-Chair)

**Cadence & process**

Biweekly (Wed 10:00 PT); consensus-driven; participation via AAIF member organisations.

**GitHub-first**

github.com/aaif/wg-observability-and-traceability

MISSION

***"Coordinate first — specify only when necessary."***

- A coordination forum aligning fragmented observability efforts, not a new standards silo
- Produces guidance only where no existing body is addressing the need
- Interoperability over innovation; adoption over theoretical elegance
- Vendor- and framework-neutral, observable-by-default

# Why this group exists

### Accountability pressure

- Production agents need auditing and evaluation: what did the agent do, why, and with what real-world effects?
- What happens across multi-modal workflows?

### A fragmented landscape

- More than a dozen competing specs and proprietary telemetry schemes (LangSmith, Arize, AgentOps, …).
- Without coordination: vendor lock-in, and traces that cannot be exchanged.

### A gap above the LLM call

- Existing telemetry standards cover individual model calls.
- Sessions, workflows, tool use and multi-agent delegation have no common representation.

**The response:** align OpenTelemetry GenAI, OWASP Agent Observability, W3C Trace Context and vendor schemes — rather than mint another competing spec.

# Scope: one trace model, five focus groups

## AGENT BEHAVIOR TRACE MODEL: The Founding Deliverable

A common data model for what an agent, did:

- Execution structure
- Sessions, turns, tool calls, hierarchy
- Reasoning and planning artifacts
- Outcomes and real-world effects
- Auditing Data

### Orchestration & Coordination

Delegation, context propagation, human-in-the-loop checkpoints

### Execution Surfaces

Tool invocation, intent = outcome attribution, retry patterns

### Protocol Observability

Normalization and trace interchange across MCP, A2A and UCP

***Direct IETF interest***

### State & Context

Memory access patterns, context assembly, evidence references

### Agentic Reasoning

Planning, evaluation criteria, self-correction patterns

### Out of scope

Products, evaluation methodology, backend prescription, inter-agent security/auth (covered by other WGs)

# Deep dive: four planes, crosswalked to the trace model

TAXONOMY — FOUR PLANES

**Orchestration plane**

Delegation, session structure, human-in-the-loop checkpoints

**Execution plane**

Tool invocation, retries, intent & outcome attribution

**Protocol plane**

MCP / A2A / UCP interchange and normalization

**Reasoning & context plane**

Plans, evaluation criteria, memory access, evidence references

**crosswalk**

*taxonomy v0.2 — in progress (#10)*

TRACE MODEL HAS THREE DIMENSIONS

**Execution structure**

Sessions, turns, tool calls, hierarchy

**Reasoning & planning**

Artifacts, criteria, self-correction

**Outcomes & effects**

Real-world results and their attribution

# Deep dive: adopt, extend, or specify?

### 1 · CANDIDATES

- OTel GenAI semantic conventions
- OWASP Agent Observability Standard
- W3C Trace Context
- Agent Trace (Cursor)
- Monocle (LF AI & Data)
- AGNTCY
- Vendor schemes

### 2 · EVALUATION FRAMEWORK

- Assessment criteria derived from charter objectives and the use-case inventory
- Inputs from the landscape analysis and the five focus groups

### 3 · GAP ANALYSIS OCT 2026

- Published alongside the Trace Model Recommendation
- The WG will not commit to a new specification before it lands (#12)

**ADOPT** an existing standard as-is

**EXTEND** an existing effort, contribute upstream (most likely OTel GenAI)

*stated bias*

**SPECIFY** a new AAIF spec, only where the gap analysis shows nothing fits

*last resort*

**Why it matters here:** if “extend” touches wire-visible propagation or cross-domain interchange, it stops being a data-model exercise and becomes protocol (potentially IETF) work.

# Current Work Timeline

| Jun 2026 | Jul 2026 | Oct 2026 | Jan 2027 |
|---|---|---|---|
| Taxonomy (draft) | Prior-work survey · use-case inventory | Gap analysis · Trace Model Recommendation | Agentic observability best practices |
| *v0.1 published  with v0.2 in progress* | *In flight: due this month* | *Planned: published deliverables* | *Planned: published* |

**IN FLIGHT ON GITHUB**

**#12** Plan the Agent Behavior Trace Model recommendation

**#10** Publish taxonomy v0.2 and trace-model crosswalk

**#9** Prioritize the use-case inventory

**#11** Establish liaison roster and cross-WG intake tracker

**#8** Onboarding checklist and focus-group entry points

**#7** Working methods and GitHub operating model

# Where this could require IETF work

### Trace propagation over HTTP

- MCP, A2A and UCP run over HTTP(S). Propagating agent session and delegation context across administrative boundaries may need header-field work beyond W3C Trace Context.

### Cross-domain trace interchange

- Interchange formats and identifiers when a trace crosses organisational boundaries.
- Negotiation and transport of trace data are protocol questions, not just data-model ones.

### Agent identity & provenance

- Tying traces to verified agent identity connects to Web Bot Auth and related IETF work.
- AAIF's Identity & Trust WG is the internal counterpart; expect liaison traffic.

### Privacy of trace data

- The charter mandates data minimisation and consent. Any wire-visible trace metadata will need RFC 6973-style privacy analysis.

### ASKS FOR IETF COMMUNITY

- **Where does protocol-level agent trace propagation belong, an existing WG, or eventually a BOF?**
- **Early IETF reviewers for the Trace Model Recommendation (Oct 2026)**
- **A liaison contact: the WG is building its liaison roster now (issue #11)**

# Discussion

Questions, pointers and pushback welcome.

- github.com/aaif/wg-observability-and-traceability
- Biweekly meetings (Wed 10:00 PT) · wg-observability-traceability@lists.aaif.io
- Help-wanted issues: onboarding (#8) · use-case inventory (#9) · liaisons (#11)

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